

Chapter 16

Agriculture — III: Cash Crops (1)

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

In a sugarcane, the sugar is stored in its

1. roots
2. stem
3. leaves
4. fruit

Answer

stem

Question 2

The content in sugarcane decreases if heavy rains continue for long time.

1. glucose
2. fructose
3. sucrose
4. lactose

Answer

sucrose

Question 3

..... are cuttings taken from old plants and planted to grow new canes.

1. Setts
2. Retts

3. Ratoons
4. Batoons

Answer

Setts

Question 4

The second or any other successive crop obtained from the roots of the leftover crop is called

1. Ratoon
2. Stalk
3. Stem
4. Cane

Answer

Ratoon

Question 5

The yield from each successive ratoon crop is than the previous one.

1. better
2. higher
3. lesser
4. worse

Answer

lesser

Question 6

Sugarcane is a exhausting crop.

1. soil
2. water
3. nitrogen
4. oxygen

Answer

soil

Question 7

Which of the following states is not a major producer of sugarcane?

1. Kerela
2. Punjab
3. Uttar Pradesh
4. Maharashtra

Answer

Kerela

Question 8

Groundnut contains per cent oil.

1. 30
2. 42
3. 50
4. 60

Answer

42

Question 9

Groundnut oil is extracted from the nuts found in the of the plant.

1. stem
2. leaves
3. roots
4. branches

Answer

roots

Question 10

Groundnut crop requires 50 to rainfall.

1. 100 cm
2. 150 cm
3. 200 cm
4. 250 cm

Answer

100 cm

Question 11

India is the largest producer of groundnuts in the world.

1. fifth
2. fourth
3. third
4. second

Answer

second

Question 12

..... is the leading producer of groundnuts in India.

1. Telangana
2. Gujarat
3. Maharashtra
4. Rajasthan

Answer

Gujarat

Question 13

..... oil is obtained from flax seeds.

1. Linseed
2. Rapeseed
3. Castor
4. Safflower

Answer

Linseed

Question 14

Soyabean has a high content.

1. calcium
2. protein
3. zinc
4. sodium

Answer

protein

Question 15

Soyabean : Kharif :: Mustard :

1. Kharif
2. Zaid
3. Rabi
4. None of the above

Answer

Rabi

Question 16

..... is the leading producer of soyabean.

1. Rajasthan
2. Maharashtra
3. Madhya Pradesh
4. Gujarat

Answer

Madhya Pradesh

Answer the following questions

Question 1(a)

Name one state in North and one in South India where sugarcane is grown extensively.

Answer

In North India, sugarcane is grown extensively in Uttar Pradesh and in South India, sugarcane is grown extensively in Tamil Nadu.

Question 1(b)

Why is the sugarcane production increasing in Maharashtra?

Answer

Sugarcane production is increasing in Maharashtra because Maharashtra has a longer harvesting season and better quality of crop owing to the favourable maritime climate free from the effects of summer loo and winter frost, sufficient irrigation and new farming techniques.

Question 1(c)

Explain briefly the Sett method of planting sugarcane.

Answer

In Sett method, new canes are usually planted by taking cuttings from the old plants. These cuttings, known as setts, quickly become established and after a few days buds sprout to form new stalks. Four to five stalks grow from each cutting. The sugarcane takes anything from 8 months to a year to mature.

Question 1(d)

What is Ratooning? State two advantages of Ratooning.

Answer

Ratooning is a method of cultivation of sugarcane. In this method, during the first harvest, the sugarcane is cut leaving a little bit of the stalk in the soil with the roots. The stalk soon puts out new shoots or ratoons. The second or any other successive crop obtained from the roots of the leftover crop is called ratoon.

Two advantages of ratooning are-

1. Crop need not be planted again, it saves labour.
2. The ratoon matures early.

Question 2(a)

How does heavy rainfall affect the quality of sugarcane?

Answer

Heavy rainfall affects the quality of sugarcane adversely as the sucrose content decreases if heavy rains continue for long.

Question 2(b)

Why does the cultivation of sugarcane require cheap labour?

Answer

The cultivation of sugarcane requires plenty of cheap labour as most of the work is done by hand. The crop is cut by hand using a long curved knife.

Question 2(c)

Why do the stalks of sugarcane need to be cut as near as possible to the ground?

Answer

The stalks of sugarcane need to be cut as near as possible to the ground because the greatest accumulation of sucrose is in the base of the stem.

Question 2(d)

State any three problems faced by sugarcane cultivators.

Answer

Three problems faced by sugarcane cultivators are-

1. Sugarcane is a soil exhausting crop and therefore, the cost of fertilizers increases the cost of production.
2. It requires high input of irrigation means.
3. The cost of transport increases the cost of production.

Question 3(a)

What extra care needs to be taken for soil health when sugarcane is cultivated?

Answer

Extra care needs to be taken for soil health when sugarcane is cultivated because sugarcane exhausts the fertility of the soil. Hence, the soil is supplemented with manures and nitrogenous fertilizers.

Question 3(b)

Mention the climate that is suitable for sugarcane cultivation.

Answer

The climate that is suitable for sugarcane cultivation is as follows-

1. **Temperature** — Sugarcane grows best in areas with 20°C to 26°C temperature. Frost is fatal for the crop. Short cool dry winter season during ripening and harvesting is ideal.
2. **Rainfall** — Sugarcane grows well in tropical regions with 100-150 cm of rainfall well distributed throughout the year. Irrigation is necessary in areas with lower rainfall or even in areas of higher rainfall during the long dry spells.

Question 3(c)

What advantage does South India have over the North with reference to sugarcane cultivation?

Answer

South India has a higher yield per hectare of sugarcane because of a longer harvesting season and better quality of crop owing to the favourable maritime climate free from the effects of summer loo and winter frost, sufficient irrigation and new farming techniques.

Question 3(d)

Mention two different ways in which sugarcane is propagated.

Answer

Two different ways in which sugarcane is propagated are-

1. Set method
2. Ratooning

Question 4(a)

What conditions of temperature and rainfall favour the growth of groundnut?

Answer

The climatic conditions that favour the growth of groundnut are-

1. **Temperature** — Groundnut requires about 20°C to 25°C temperature. Dry weather is needed during the time of ripening.

2. **Rainfall** — It requires light to moderate rainfall between 50 to 100 cm. Rainfall should be well distributed.

Question 4(b)

Which two states in India are the leading producers of groundnut? Why is groundnut mostly grown in Peninsular India?

Answer

Gujarat and Tamil Nadu in India are the leading producers of groundnut.

Groundnut is mostly grown in Peninsular India because the red, yellow and black cotton soils of Peninsular India suit it well. Such light soils are essential as the nut ripens in the soil.

Question 4(c)

Give three uses of groundnut.

Answer

Three uses of groundnut are-

1. It is used for the manufacture of hydrogenated oil.
2. It is used for making margarine, medical emulsions and soap.
3. Groundnut oil is used as a cooking medium and the nuts are eaten raw, roasted, salted or sweetened.

Question 4(d)

What is the residue after crushing oilseeds called? Give two of its uses.

Answer

The residue after crushing oilseeds is called oilcake.

Oilcake is used as a fodder for animals and also serves a good manure in the fields.

Question 5(a)

Name any two oilseeds grown in India. Which oilseed is grown as a rainfed crop?

Answer

Two oilseeds grown in India are groundnut and linseed.

Groundnut is grown as a rainfed crop.

Question 5(b)

State the conditions of temperature and rainfall necessary for the growth of mustard.

Answer

The climatic conditions necessary for the growth of mustard are-

1. **Temperature** — Mustard grows best in cool climate. Ideal temperature ranges from 10°C to 20°C.
2. **Rainfall** — The crop requires rainfall varying between 25 to 40 cm.

Question 5(c)

(i) Which soil type favours the growth of mustard? Name an area where it grows.

(ii) Mention two uses of mustard.

Answer

(i) Mustard can be grown on loams but slightly heavier soils are preferred.

Mustard is widely grown in the Sutlej-Ganga plain.

(ii) Two uses of mustard are-

1. Its seeds are crushed for edible oil.
2. The oilcake of mustard is an important cattle feed.

Question 5(d)

Mention the climatic conditions necessary to grow soya bean. State two uses of soya bean.

Answer

The climatic conditions necessary to grow soya bean are-

1. **Temperature** — Soyabean requires temperature in the range of 13°C to 24°C. Temperature above 13°C is necessary for the normal growth of soyabean.
2. **Rainfall** — It grows in regions with 40 cm to 60 cm rainfall. The rainfall should be well distributed throughout the growing season.

Two uses of Soyabean are-

1. It is eaten as vegetable or made into soya sauce.

2. Due to its high protein content, it is consumed as soya milk or tofu (cheese) as a substitute for animal protein.

Thinking Skills

Question 1

Imagine you are a farmer living in Bihar and you want to grow a cash crop on your farm. Which one you will choose and why?

Answer

As a farmer living in Bihar, if I were to choose a cash crop for my farm, I would select mustard because of the following reasons:

1. Agro-climatic Suitability — Mustard is well-suited to the agro-climatic conditions of Bihar, including its diverse climate with hot summers and cold winters.
2. Market Demand — Mustard oil extracted from mustard seeds is widely used for cooking purposes, ensuring a consistent market demand and potential profitability.
3. Crop Rotation Benefits — Mustard fits well into crop rotation systems, helping suppress weeds, pests, and diseases.
4. Diversification — Cultivating mustard allows for diversification of income sources on the farm, reducing dependence on a single crop and mitigating risks associated with market fluctuations.

Question 2

You live in Uttar Pradesh and your cousin lives in Madhya Pradesh. Tell the benefits of an edible oil seed grown in your state and get the information about an oilseed that is extensively grown in your cousin's state.

Answer

As a resident of Uttar Pradesh, one of the significant oilseeds grown in the state is mustard. Mustard cultivation offers several benefits:

1. Multiple Uses — Mustard oil extracted from mustard seeds is commonly used for cooking purposes in many households. Additionally, mustard seeds are used in pickles, condiments, and spice blends, adding flavour to various dishes.
2. Nutritional Value — Mustard seeds are rich in essential nutrients like omega-3 fatty acids, protein, fiber, vitamins, and minerals, which contribute to a healthy diet.
3. Adaptability — Mustard is a versatile crop that can be grown in various agro-climatic conditions.

4. Crop Rotation Benefits — Mustard is an excellent choice for crop rotation due to its ability to suppress weeds and diseases.
5. Income Generation — Mustard farming can generate income for farmers due to consistent market demand.

In Madhya Pradesh, one extensively grown oilseed is soyabean. Some of the benefits of soyabean cultivation include:

1. High Protein Content — Soyabean is known for its high protein content.
2. Edible Oil Production — Soyabean seeds are processed to extract soyabean oil, widely used in cooking and food industries.
3. Export Potential — Madhya Pradesh's significant production of soyabean contributes to India's overall export of soyabean and soyabean products.
4. Soil Improvement — Soyabean cultivation improves soil quality through nitrogen fixation.
5. Livelihood Support — Soyabean farming supports livelihoods for farmers and related industries, benefiting the rural economy.